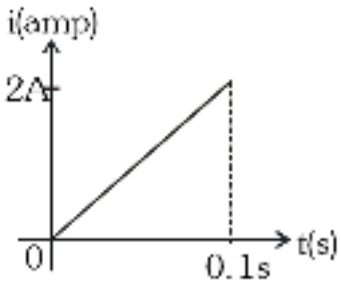
PHYSICS

1)

In a coil of resistance 20 Ω, the induced current developed by changing magnetic flux through it, is shown in figure as a function of time. The magnitude of change in flux through the coil is :-



(1) 6 Wb

(2) 4 Wb

(3) 8 Wb

(4) 2 Wb

2)

Match the following columns :

Column-I		Column-II	
A.	V_R	1.	$\frac{\pi}{2}$ ahead of I
B.	V_C	2.	Parallel to I
C.	V_L	3.	$\frac{\pi}{2}$ behind I

Codes :-

	A	B	C
(1)	1	2	3
(2)	2	3	1
(3)	3	2	1
(4)	1	3	2

(1) 1

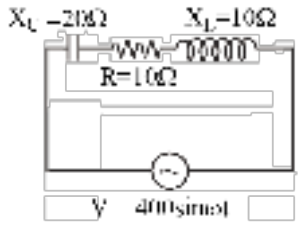
(2) 2

(3) 3

(4) 4

3)

In the LCR circuit shown in figure :-



(1) Current will leads the voltage

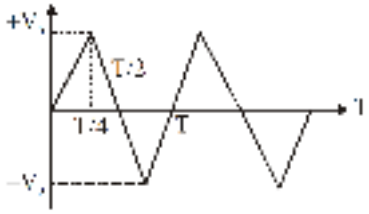
(2) rms value of currents is 20 A

(3) Power factor of the circuit is $\frac{1}{\sqrt{2}}$

(4) All of the above

4)

The voltage time (V-t) graph for triangular wave having peak value V_0 is as shown in fig.



The rms value of V in time interval from t = 0 to T/4 is

(1) $\frac{V_0}{\sqrt{3}}$

(2) $\frac{V_0}{2}$

(3) $\frac{V_0}{\sqrt{2}}$

(4) None of these

5) The resonant frequency of an RLC circuit is 125000 cycles per second. Capacitor C is replaced by another identical capacitor with a dielectric medium of constant K . In this case, the frequency decreases by 25 KHz. The value of K is :-

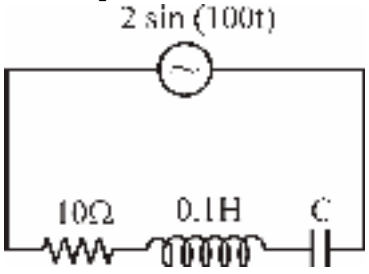
(1) 3

(2) 2.1

(3) 1.56

(4) 1.7

6) The power factor of circuit is $1/\sqrt{2}$. The capacitive reactance of the circuit is equal to :-



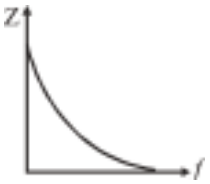
(1) 2 Ω

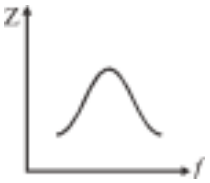
(2) Zero

(3) 1 Ω

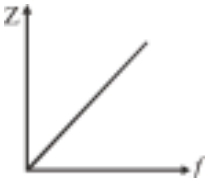
(4) Infinite

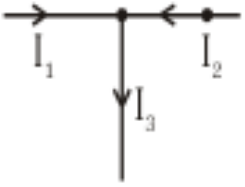
7) Which one of the following curves represents the variation of impedence (Z) with frequency f in series LCR circuit :-

(1) 

(2) 

(3) 

(4) 



8) If current I_1 = 3A sin ωt and I_2 = 4A cos ωt, then I_3 is :-

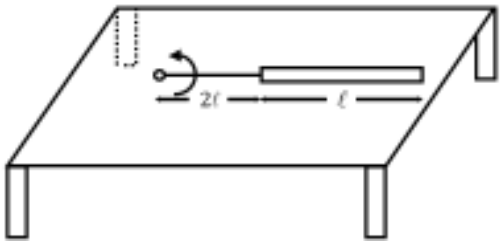
(1) 5A sin (ωt + 53°)

(2) 5A sin (ωt + 37°)

(3) 5A sin (ωt + 45°)

(4) 5A sin (ωt + 30°)

9) A metallic rod of lenght '□' is tied to a string of length 2□ and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field 'B' in the region, the emf induced across the ends of the rod is:



(1) $\frac{4B\omega\ell^2}{2}$

(2) $\frac{5B\omega\ell^2}{2}$

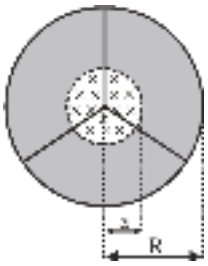
(3) $\frac{2B\omega\ell^2}{2}$

(4) $\frac{3B\omega\ell^2}{2}$

10) A line charge λ per unit length is lodged uniformly onto the rim of a wheel of mass M and radius R. The wheel has light nonconducting spokes and is free to rotate without friction about its axis see figure. A uniform magnetic field extends over a circular region within the rim. It is given by,

$\vec{B} = -B_0\hat{k}$ $(r \le a\ ;\ a < R)$

$= 0$ (otherwise)



What is the angular velocity of the wheel after the field is suddenly switched off ?

(1) $-\frac{B_0\pi a^2\lambda}{MR}\hat{k}$

(2) $\frac{B_0\pi a^2\lambda}{MR}\hat{k}$

(3) $-\frac{B_0\pi R^2\lambda}{Ma}\hat{k}$

(4) $-\frac{B_0\pi a^2\hat{k}}{2MR}$

11) In an AC circuit, voltage $V = V_0\sin\omega t$ and inductor L is connected across the circuit. Then the instantaneous power will be :-

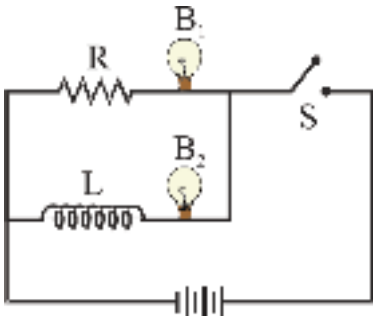
(1) $\frac{V_0^2}{2\omega L}\sin\omega t$

(2) $\frac{-V_0^2}{2\omega L}\sin\omega t$

(3) $\frac{-V_0^2}{2\omega L}\sin 2\omega t$

(4) $\frac{V_0^2}{\omega L}\sin 2\omega t$

12) Figure shows two bulbs B_1 and B_2, resistor R and an inductor L. When the switch S is turned off :



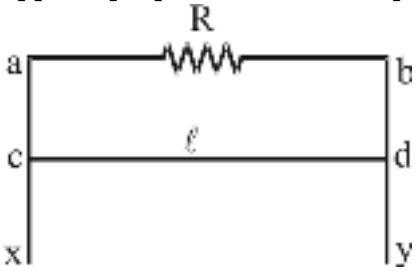
(1) Both B_1 and B_2 die out promptly

(2) Both B_1 and B_2 die out with some delay

(3) B_1 dies out promptly but B_2 with some delay

(4) B_2 dies out promptly but B_1 with some delay

13) A wire cd of length and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b. A uniform magnetic field B is applied perpendicular to the plane abcd such that cd moves with a constant velocity given by :-



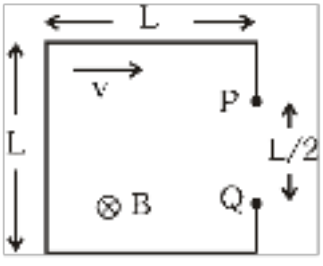
(1) $\frac{mgR}{B\ell}$

(2) $\frac{mgR}{B^2\ell^2}$

(3) $\frac{mgR}{B^3\ell^3}$

(4) $\frac{mgR}{B^2\ell}$

14) The loop shown moves with a constant velocity ‘v’ in a uniform magnetic field of magnitude ‘B’ directed into the paper. The potential difference between P and Q is ‘e’:-

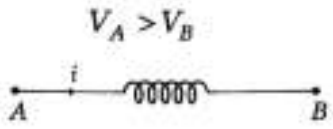


(1) $e = \frac{BLv}{2}$, Q is positive with respect to P

(2) $e = \frac{BLv}{2}$, P is positive with respect to Q

(3) e = 0

(4) e = BLv, Q is positive with respect to P

15) **Assertion :** If current shown in the figure is increasing then 

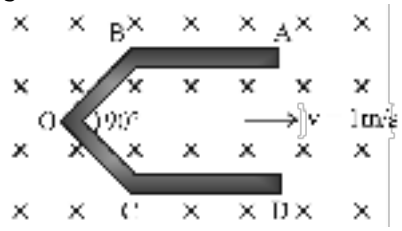
Reason : If current passing through an inductor is constant, then both ends of the inductor are at same potential.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are true and but Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

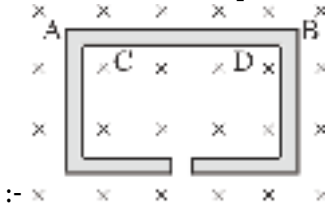
16) A conductor ABOCD moves along its bisector with a velocity of 1 m/s through a perpendicular magnetic field of 1 wb/m^2, as shown in figure. If all the four sides are of 1 m length each, then the induced emf between point A and D is :- 

(1) 0

(2) 1.41 volt

(3) 0.71 volt

(4) None of these

17) A wire is bent to form the double loop shown in figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, current will flow from 

:-

(1) A to B and C to D

(2) B to A and D to C

(3) A to B and D to C

(4) B to A and C to D

18) A transformer is used to light a 100 W and 110V lamp from a 220 V mains. If the main current is 0.5 amp, the efficiency of the transformer is approximately :-

(1) 10%

(2) 30%

(3) 50%

(4) 90%

19) A uniform magnetic field exists in region given by $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. A rod of length 5m is placed

along y - axis is moved along x - axis with constant speed 1 m/sec. Then induced e.m.f. in the rod will be:

(1) zero

(2) 25 volt

(3) 20 volt

(4) 15 volt

20) A hollow cylinderical wire carries a current I, having inner and outer radii 'R' and 2R respectively. Magnetic field at a point which 3R/2 distance away from its axis is :-

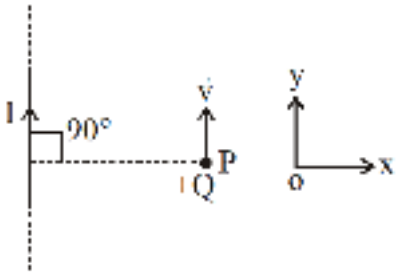
(1) $\frac{5\mu_0 I}{18\pi R}$

(2) $\frac{\mu_0 I}{36\pi R}$

(3) $\frac{5\mu_0 I}{36\pi R}$

(4) $\frac{5\mu_0 I}{9\pi R}$

21) A long straight wire carries a current I. At the instant, when a charge +Q at point P has velocity $\vec{v}$ as shown in figure. The direction of magnetic force on the charge is :-



(1) along ox

(2) along oy

(3) opposite to ox

(4) opposite to oy

22) A long straight wire in the horizontal plane carries a current of 50 A in north to south direction. Give the magnitude and direction of B at a point 2.5m east of the wire :-

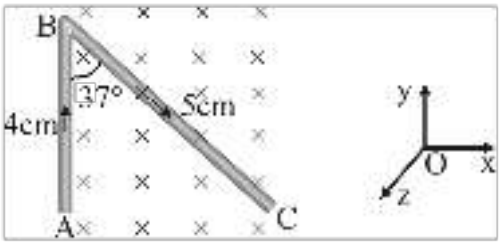
(1) 4×10^{-6} T upwards

(2) 6×10^{-6} T down

(3) 4×10^{-6} T east

(4) Both (1) and (3)

23) A uniform conducting wire ABC has a mass of 10 g. A current of 2A flows through it. The wire is kept in a uniform magnetic field B = 2T. The acceleration of the wire will be :-



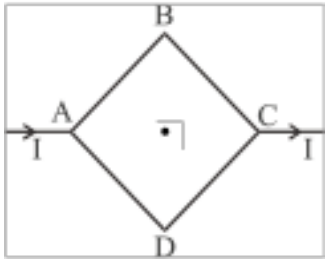
(1) Zero

(2) 12 ms^{-2} along y-axis

(3) 1.2×10^{-3} ms^{-2} along y-axis

(4) 0.6×10^{-3} ms^{-2} along y-axis

24) A square loop with each side d is shown in the figure. The resistance of the wire ABC is r while that of ADC is 2r. Then the magnetic field at the centre of the loop, when the wires are uniform is :



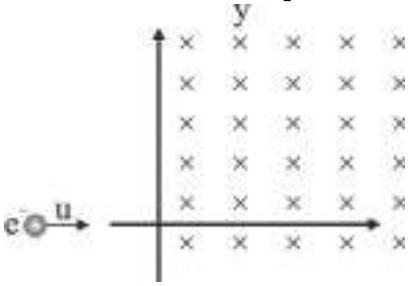
(1) $\frac{\sqrt{2}\mu_0 I}{\pi d}$

(2) $\frac{\sqrt{2}\mu_0 I}{3\pi d}$

(3) $\frac{2\sqrt{2}\mu_0 I}{3\pi d}$

(4) Zero

25) An electron moving with a speed u along the positive x-axis at y = 0 enters a region of uniform magnetic field $\vec{B} = -B_0\hat{k}$ which exists to the right of y-axis. The electron exits from the region after some time with the speed v at co-ordinate y.



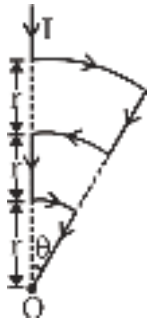
(1) v > u, y < 0

(2) v = u, y > 0

(3) v > u, y > 0

(4) v = u, y < 0

26) Shown in the figure is a conductor carrying a current I. The magnetic field intensity at the point



O (common centre of all the three arcs) is :-

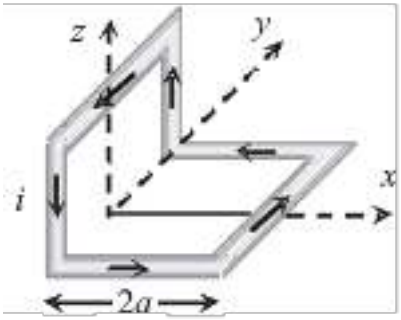
(1) $\frac{5\mu_0 I\theta}{24\pi r}$

(2) $\frac{\mu_0 I\theta}{24\pi r}$

(3) $\frac{11\mu_0 I\theta}{24\pi r}$

(4) zero

27) A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P



(a,0,a) points in the direction

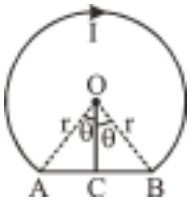
(1) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$

(2) $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$

(3) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(4) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$

28) A wire is bent in the form of a circular arc with a straight portion AB. Magnetic induction at O



when current I flowing in the wire, is :-

(1) $\frac{\mu_0}{2r}(\pi-\theta+\tan\theta)$

(2) $\frac{\mu_0 I}{2\pi r}(\pi+\theta-\tan\theta)$

(3) $\frac{\mu_0 I}{2\pi r}(\pi - \theta + \tan\theta)$

(4) $\frac{\mu_0 I}{2\pi r}(-\tan\theta + \pi - \theta)$

29) The coercivity of a small magnet where the ferromagnet gets demagnetized is 3×10^3 A m^{-1}. The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is :

(1) 3A

(2) 6 A

(3) 30 mA

(4) 60 mA

30) Match the column :

	A		B
(a)	$-1 \leq \chi_m < 0$	(p)	Dia-magnetic
(b)	$0 < \chi_m < 1$	(q)	Ferromagnetic
(c)	$\chi_m >> 1$	(r)	Paramagnetic
(d)	$\mu > \mu_0$		

(1) a → p, b → r, c → q, d → q, r

(2) a → p, b → r, c → q, d → p, q

(3) a → p, b → r, c → q, d → p

(4) a → r, b → p, c → q, d → p, r

31) An iron rod of cross-sectional area 4 sq cm is placed with its length parallel to a magnetic field of intensity 1600 amp/m. The flux through the rod is 4×10^{-4} weber. What is the permeability (In weber/amp-m) of the material of the rod :

(1) 0.625

(2) 6.25

(3) 0.625×10^{-3}

(4) None of these

32) A disc of radius 'R' and mass 'm' has charge 'q' uniformly distributed on it. It rotates with frequency f cycle/sec about it geometric axis. The ratio of magnetic moment and moment of inertia is -

(1) $\frac{q\pi f}{2m}$

(2) $\frac{q\pi f}{m}$

(3) $\frac{q(2\pi f)}{mR^2}$

(4) $\frac{q\pi f}{mR^2}$

33) An electron is revolving around a proton in a circular orbit of diameter 1Å. If it produces a magnetic field of 14 wb/m^2 at the proton, then its angular velocity will be about :-

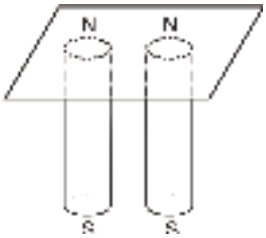
(1) 4.375×10^{16} rad/s

(2) 2.25×10^{14} rad/s

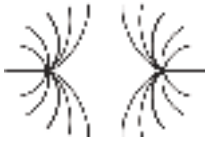
(3) 4×10^{15} rad/s

(4) 8.75×10^{16} rad/s

34) Two long, identical bar magnets are placed under a horizontal piece of paper, as shown in figure. The paper is covered with iron filings. When the two north poles are at small distance apart and touching the paper, the iron filings spread into a pattern that shows the magnetic lines of forces. Which of the following best illustrates the pattern that results ?



(1) 

(2) 

(3) 

(4) 

35) ***Assertion (A) :*** A linear solenoid carrying current is equivalent to a bar magnet. ***Reason (R) :*** The magnetic field lines due to current carrying solenoid resemble with those of a bar magnet.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is incorrect but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

36) The current in the windings of a toroid is 2.0A. There are 400 turns and the mean circumferential length is 40cm. If the inside magnetic field is 1.0T, the relative permeability is near to :-

(1) 100

(2) 200

(3) 300

(4) 400

37) In order to establish an instantaneous displacement current of 1mA in the space between the plates of 2μF parallel plate capacitor, the rate of change of potential difference is

(1) 100 Vs^{-1}

(2) 200 Vs^{-1}

(3) 300 Vs^{-1}

(4) 500 Vs^{-1}

38) An electromagnetic wave propagating along North has its electric field vector upwards. Its magnetic field vector point towards

(1) North

(2) East

(3) West

(4) Downwards

39) The part of the spectrum of the electromagnetic radiation used to cook food is

(1) ultraviolet rays

(2) cosmic rays

(3) X-rays

(4) microwaves

40) **Statement-I:** The energy contained in a small volume through which an em wave is passing oscillates with the frequency of the wave.

Statement-II: Energy density of the wave is given by $\frac{1}{2}\varepsilon^{0}E^{2}$.

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Both Statement I and Statement II are incorrect.

(4) Statement I is incorrect but Statement II is correct.

41) 50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on $1m^2$ surface area is $x \times 10^{-8}$ N, where the value of x is approximately.

(1) 21

(2) 10

(3) 11

(4) 42

42) A radiation is emitted by 1000 W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2 m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1}$ V/m. Value of x is _____ . (Rounded off to the nearest integer)

[Take $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2N^{-1}m^{-2}$,
$c = 3 \times 10^8\ ms^{-1}$]

(1) 137

(2) 187

(3) 25

(4) 20

43) In an experiment to find the resistance of galvanometer by half deflection method, a 5V battery and a high resistance of 4.9 kΩ are connected in circuit. In absence of any shunt resistance galvanometers reads 20 divisions when current flows in circuit. To reduce the deflection by half the value of shunt resistance used is 98Ω. The figure of merit of galvanometer is given as :

(1) 5μA / division

(2) 20μA / division

(3) 25μA / division

(4) 50μA / division

44) A proton and an alpha particle are separately projected in a region where a uniform magnetic field exists. The initial velocities are perpendicular to the direction of magnetic field. If both the particles move around magnetic field in circles of equal radii, the ratio of momentum of proton to alpha particle $\left(\frac{P_p}{P_\alpha}\right)$ is :-

(1) 1

(2) $\frac{1}{2}$

(3) 2

(4) $\frac{1}{4}$

45) A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3}J$ of work to turn it through 60°. The torque needed to maintain the needle in this position will be

(1) $2\sqrt{3}$ N-m

(2) 3 N-m

(3) $\sqrt{3}$ N-m

(4) 3/2 N-m

CHEMISTRY

1) $\underset{\text{(Yellow solution)}}{X} \xrightarrow{(CH_3COO)_2Pb} \underset{\text{(Yellow ppt.)}}{Y}$

Then X and Y are :

(1) X = AgCl, Y = CH_3COOAg

(2) X = Na_2CrO_4, Y= $PbCrO_4$

(3) X = NaBr, Y = PbBr2

(4) X = $FeCl_3$, Y = $PbCl_2$

2) Which metal is present in brass, bronze and german silver :

(1) Zn

(2) Mg

(3) Cu

(4) All

3) Colour of borax bead test of Fe in oxidising flame is :

(1) Red

(2) Yellow

(3) Green

(4) White

4) Ni^{+2} ions are tested by :

(1) Methyl orange

(2) $EDTA^{-4}$

(3) $DMG^{\ominus}$

(4) NaOH

5) Which compound does not exist ?

(1) MnF_6

(2) MnF_4

(3) MnO_3F

(4) CrF_6

6) Match the following :

	Column-I (ppt)		**Column-II (Colour)**
(A)	$PbCl_2$	(P)	Black
(B)	$BaCrO_4$	(Q)	White
(C)	MnS	(R)	Yellow
(D)	NiS	(S)	Pink

Correct option is :

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-P, C-S, D-R

(3) A-Q, B-R, C-S, D-P

(4) A-Q, B-R, C-P, D-S

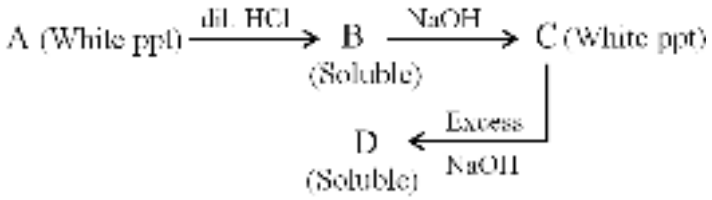
7) Wrong statement about solution containing Fe^{+3} :

(1) can reduced by $SnCl_2$

(2) gives ppt with NaOH

(3) gives blue ppt with $K_4[Fe(CN)_6]$

(4) gives blue ppt with KSCN



8)

'A' can be :

(1) $Fe(OH)_3$

(2) $Cu(OH)_2$

(3) ZnS

(4) $CoCl_2$

9) Group reagent for group IV cation is :

(1) dil. H_2SO_4

(2) H_2S in presence of dil. HCl

(3) H_2S in presence of NH_4OH

(4) $NH_4OH + NH_4Cl + (NH_4)_2CO_3$

10) Incorrect match is :

(1) Tetra ethyl lead – σ-bonded OMC

(2) Ferrocene – π- bonded OMC

(3) $[Ni(CO)_4]$ – σ and π bonded OMC

(4) $[Ti(OC_6H_5)_4]$ – σ-bonded OMC

11) Incorrect match is :

	Identification test	**Colour**
(1)	Starch-iodide test	Blue
(2)	Chromyl chloride test	Green
(3)	Layer test of bromide	Red-brown
(4)	Layer test of iodide	Violet

(1) 1

(2) 2

(3) 3

(4) 4

12) In acidic solution of sodium nitroprusside when H_2S gas is passed then

(1) Violet colour solution is formed

(2) Violet ppt is formed

(3) H_2S will form S

(4) No colour change is observed

13) $Cu^{+2} + K_4[Fe(CN)_6] \rightarrow$ Complex A
What is formula and colour of complex A ?

(1) $Fe_4[Fe(CN)_6]_3$, Prussian blue

(2) $[Cu(NH_3)_4]SO_4$, Deep blue

(3) Both (1) and (2)

(4) $Cu_2[Fe(CN)_6]$, Chocolate brown ppt

14) Which of the following will have only two stereoisomeric forms ?
(I) $[Co\ (NO_3)_3\ (NH_3)_3]$
(II) $K_3[Fe(OX)_3]$
(III) $[CoCl_2(en)_2]^+$
(IV) $[CoBrCl(OX)_2]^{3-}$

(1) I only

(2) I and II

(3) III and IV

(4) All

15) Disodium hydrogen phosphate is used to test :

(1) Mg^{+2}

(2) Be^{+2}

(3) Ca^{+2}

(4) Ni^{+2}

16) Before adding the reagent of group III, if some drops of conc. HNO_3 is not added then :

(1) Ppt of $Fe(OH)_3$ will be formed easily

(2) Sn^{+2} will not be oxidised to Sn^{+4}

(3) Fe^{+2} will not be oxidised to Fe^{+3}

(4) Ppt of $Al(OH)_3$ will be formed

17) Which is incorrect statement ?

(1) Hg_2Cl_2 does not give chromyl chloride test

(2) H_2S will not give purple coloured complex with sodium nitroprusside

(3) $CuSO_4$ gives deep blue soluble complex with excess NH_4OH solution

(4) NH_3 gas gives blue colour ppt with Nessler reagent

18) Match the list-I with list-II.

	List-I (Cation)		**List-II (Group reagent)**
A	Pb^{+2}, Cu^{+2}	(i)	H_2S + HCl
B	Al^{+3}, Fe^{+3}	(ii)	$(NH_4)_2CO_3$ + NH_4OH
C	Co^{+2}, Ni^{+2}	(iii)	NH_4OH + NH_4Cl
D	Ba^{+2}, Ca^{+2}	(iv)	H_2S + NH_4OH

Correct match is :

(1) A-iv ; B-ii ; C-iii ; D-i

(2) A-i ; B-iii ; C-iv ; D-ii

(3) A-iii ; B-i ; C-iv ; D-ii

(4) A-i ; B-iii ; C-ii ; D-iv

19) **Assertion :** Test of NH_4^+ cannot be done within group analysis.
Reason : During group analysis, several times NH_4^+ compounds are added at the different steps.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

20) Thenard blue is :

(1) CoO.ZnO

(2) $CoO.Al_2O_3$

(3) CoO.MgO

(4) $CuSO_4.5H_2O$

21) Which of the following octet rule ?

(1) $BeH_2(s)$

(2) I_2Cl_6

(3) $BeCl_2(s)$

(4) B_2H_6

22) Which of the following is a cyclic oxy acid ?

(1) $H_4P_2O_7$

(2) $H_4P_2O_6$

(3) $H_4P_2O_5$

(4) $H_3P_3O_9$

23)

The purest and thermodynamically most stable allotropes of carbon are respectively :

(1) Diamond, graphite

(2) Fullerenes, graphite

(3) Graphite, diamond

(4) Graphite, fullerenes

24) Which of the following order is correct ?
(I) $CO_2 < SiO_2 < GeO_2 < SnO_2$ (Oxidising nature)
(II) $MnO_4^- < TcO_4^- < ReO_4^-$ (Oxidising nature)
(III) $CH_4 < SiH_4 < GeH_4 < SnH_4$ (Reducing nature)

(1) I, II, III

(2) II, III

(3) I only

(4) I, III

25) Which of the following property of hydrides of group-15 increases on moving down the group ?

(1) Thermal stability

(2) Bond angle

(3) Reducing nature

(4) Lewis basic strength

26) Which of the following statement is wrong ?

(1) Single N-N bond is stronger than the single P-P bond

(2) NO_2 is paramagnetic in nature

(3) PH_3 can act as ligand in the formation of coordination compound with transition elements

(4) Covalency of nitrogen in N_2O_4 is 4

27) $Br_2 + 2X^- \rightarrow 2Br^- + X_2$
$Cl_2 + 2Y^- \rightarrow 2Cl^- + Y_2$
The incorrect match regarding above reaction is :

(1) $X_2 = I_2$

(2) $Y_2 = I_2$

(3) $X_2 = Cl_2$

(4) $Y_2 = Br_2$

28) XeF_4 act as fluoride acceptor with :

(1) PF_5
(2) SbF_5
(3) KF
(4) All of these

29) Formula of potassiumdicyanidodioxalato nickelate (II) is :

(1) $K_4[Ni(CN)(OX)_2]$
(2) $K_3[Ni(CN)_2(OX)]$
(3) $K_4[Ni(CN)_2(OX)_2]$
(4) $K_2[Ni(CN)_2(OX)_2]$

30) In which metal carbonyl (C–O) bond order is minimum ?

(1) $[Mn(CO)_6]^+$
(2) $[Fe(CO)_5]$
(3) $[Cr(CO)_6]$
(4) $[V(CO)_6]^-$

31) **Assertion :** Boron does not form B^{+3} ion.
Reason : Boron does not form compound in its ground state.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

32) When 0.1 mole of $CoCl_3.5NH_3$ is treated with excess of $AgNO_3$, 0.2 moles of AgCl are obtained. Then the conductivity of solution will correspond :

(1) 1 : 3 electrolyte
(2) 1 : 2 electrolyte
(3) 1 : 1 electrolyte
(4) 3 : 1 electrolyte

33) Which of the following is not an inner orbital complex ?

(1) $[Co(OX)_3]^{-3}$
(2) $[Co(CN)_6]^{-3}$
(3) $[Fe(CN)_6]^{-4}$
(4) $[Ni(NH_3)_6]^{+2}$

34) Crystal field stabilisation energy for high spin d^4 octahedral complex is :

(1) $-0.6\Delta_0$

(2) $-1.8\Delta_0$

(3) $-1.6\Delta_0 + p$

(4) $-1.2\Delta_0$

35) Which one is not coloured ?

(1) $Na_2[CdCl_4]$

(2) $K_3[Fe(CN)_6]$

(3) $Na_2[CuCl_4]$

(4) $K_3[Cr(CN)_6]$

36) **Assertion :** Cr^{+2} is stronger reducing agent than Fe^{+2}.
Reason : $E^{o}_{M^{3+}/M^{+2}}$ are negative for both Fe and Cr.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

37) (X) + K_2CO_3 + air $\xrightarrow{\text{Heat}}$ (Y)
(Y) + $Cl_2 \rightarrow$ (Z) purple
Which of the following is correct ?

(1) X = Black, MnO_2, Y = Blue, K_2CrO_4, Z = $KMnO_4$

(2) X = Green, Cr_2O_3, Y = Yellow, K_2CrO_4, Z = $K_2Cr_2O_7$

(3) X = Black, MnO_2, Y = Green, K_2MnO_4, Z = $KMnO_4$

(4) X = Black, Bi_2O_3, Y = Colourless, $KBiO_2$, Z = $KBiO_3$

38) Which among the following pair is interstitial compounds ?

(1) SiH_4, LiH

(2) Fe_3C, Mn_4N

(3) Al_4C_3, SiC

(4) $TiCl_4$, MnO_2

39) Which of the following compound is used for the estimation of CO ?

(1) I_2O_5

(2) KI + H_2O

(3) I_2

(4) Cl_2O_6

40) The hybridisation and geometry of $[Fe(CO)_4]^{-2}$ is :

(1) sp^3d, TBP
(2) sp^3, tetrahedral
(3) dsp^2, square planar
(4) sp^3, TBP

41) Which of following statement is false :

(1) SF_6 is gas at room temperature.
(2) SeF_6 is liquid at room temperature.
(3) TeO_2 is oxidising in nature.
(4) SeF_4 is liquid at room temperature.

42) For which of the following complexes $\Delta_0 > P$?

(1) $[Ir(H_2O)_6]^{+3}$
(2) $[Fe(NH_3)_6]^{+2}$
(3) $[Mn(NH_3)_6]^{+2}$
(4) $[Co(H_2O)_6]^{+2}$

43) Which of the following fluoride does not exist ?

(1) FeF_3
(2) BiF_5
(3) CuF
(4) CoF_3

44) Which noble gas is most soluble in water ?

(1) He
(2) Ne
(3) Ar
(4) Xe

45) In which reaction no colour change will be observed ?

(1) $K_2Cr_2O_7 \xrightarrow{CO_2}$
(2) $Na_2CrO_4 \xrightarrow{CO_2}$
(3) $K_2Cr_2O_7 \xrightarrow{SO_2}$
(4) $Na_2S \xrightarrow{Na_2[Fe(CN)_5(NO)]}$

BIOLOGY

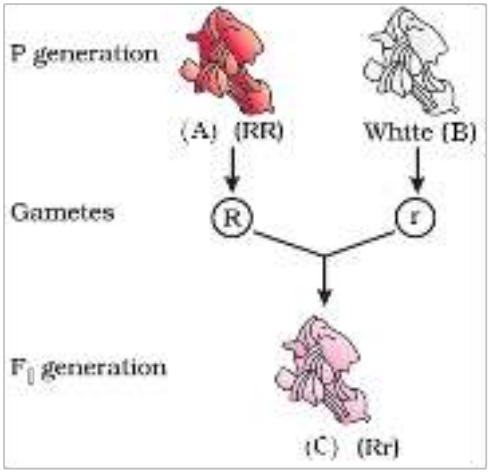
1) In ABO blood grouping how many genotypes show co-dominance ?

(1) 2
(2) 1
(3) 6
(4) 3

2) Alleles of different genes that are on the same chromosome can occasionally be separated by a phenomenon called -

(1) Independent assortment
(2) Crossing over
(3) Disjunction
(4) Linkage

3) Parent generation :



Correctly identify A, B and C in given cross about inheritance of flower colour in snapdragon:

(1) A - Red, B - rr, C - Pink
(2) A - Pink, B - Rr, C - Red
(3) A - Red, B - Rr, C - Pink
(4) A - Pink, B - rr, C - Red

4) An impure green pod garden pea plant is crossed with pure yellow pod plant then find out the phenotypic ratio?

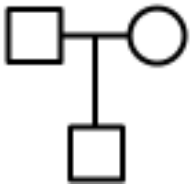
(1) 3 : 1
(2) 1 : 1 : 1 : 1
(3) 1 : 1
(4) 1 : 2 : 1

5) Which of the following represent consanguineous marriage?

(1) 

(2) 

(3) 

(4) 

6) Based upon observations on dihybrids crosses Mendel proposed a second set of generalisations that we called -

(1) Mendel's Law of Dominance
(2) Mendel's law of purity of gametes.
(3) Mendel's law of independent assortment
(4) Mendel's law of segregation

7) What happens when Glutamic acid is substituted by Valine at the sixth position of the beta globin chain of the haemoglobin molecule ?

(1) The shape of the RBC changes from elongated to biconcave disc.
(2) Under low oxygen tension the shape of the RBC changes from biconcave disc to elongated sickle like structure.
(3) The shape of the RBC remains the same.
(4) The RBC will carry more oxygen.

8) Which of the following is not an example of male heterogamety ?

(1) Grasshopper
(2) Human
(3) *Drosophila*
(4) Birds

9) A colourblind man has a colour blind sister but a normal brother. The phenotype of his parents are :-

(1) Father colour blind and mother normal
(2) Father normal and mother colourblind
(3) Both parents are normal
(4) Both parents are colourblind

10) What is allele ?

(1) True breeding homozygous

(2) Heterozygous

(3) Slightly different form of same gene

(4) Different phenotype

11) Read the following statements carefully and choose the **correct** option :-
(a) Phenotype is the external manifestation of gene product.
(b) Individuals with similar phenotypes may not belong to same genotype.
(c) Individuals with different phenotypes usually have similar genotypes.
(d) Phenotype is not influenced by environment.
(e) Phenotype always remains same throughout the life of an individual.

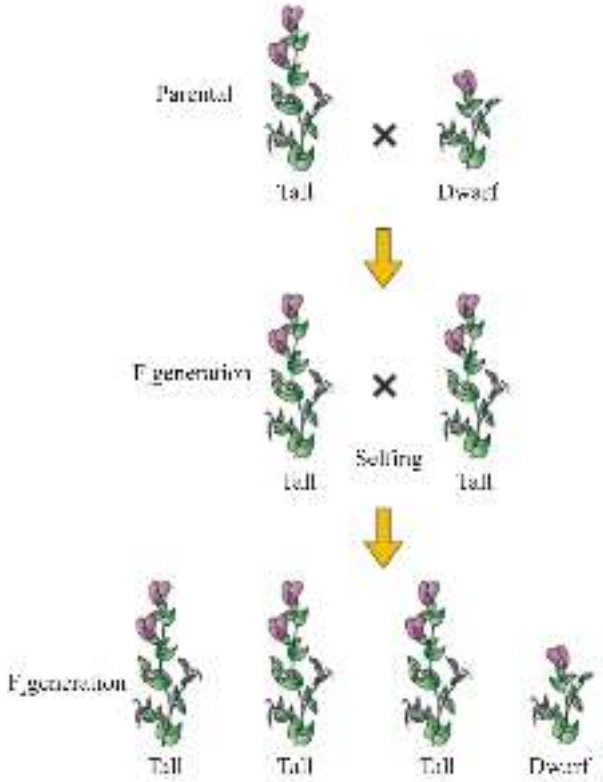
(1) b, c & d are correct

(2) a, b & e are correct

(3) c & d are correct

(4) a & b are correct

12) Given below is the diagrammatic representation of a monohybrid cross. In this, find out the genotypes of tall plants in all the given generations :-



	A Parents	**B** F_1-generation	**C** F_2-generation
(1)	TT	TT,Tt	TT,Tt
(2)	Tt	Tt	TT,Tt
(3)	TT	Tt	TT,Tt
(4)	TT	TT	Tt

(1) 1

(2) 2

(3) 3

(4) 4

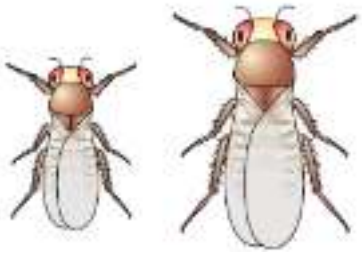
13) Consider following genetic disorders:
Haemophilia, Sickle cell anaemia, Cystic fibrosis, Thalassemia, Phenylketonuria, Colour blindness, Klinefelter syndrome, Turner's syndroms.
How many of the above disorders are mendelian disorders?

(1) 6

(2) 2

(3) 8

(4) 3

14) Given below is, the figure of *Drosophila melanogaster*.  It was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan, because.

(1) A single mating produces two young flies.

(2) Smaller female is easily recognisable from larger male.

(3) It completes life cycle in about two weeks.

(4) It reproduces parthenogenetically only.

15) Match the following :-

(A)	Human female	(i)	2A + XY
(B)	Male *Drosophila*	(ii)	2A + XX
(C)	Male Grasshopper	(iii)	2A + ZW
(D)	Female Birds	(iv)	2A + XO

	A	B	C	D
(1)	(i)	(ii)	(iv)	(iii)
(2)	(ii)	(i)	(iii)	(iv)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(i)	(ii)	(iii)	(iv)

(1) 1

(2) 2

(3) 3

(4) 4

16) **Assertion:** The contrasting traits studied by Mendel did not show any blending at either F_1 or F_2 stage.
Reason: Only one parental traits was expressed in the F_1 generation while at the F_2 stage both the traits were expressed in the proportion 3:1.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

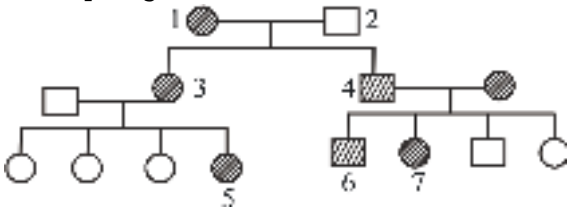
17) **Statement I** : Human skin colour is controlled by three polygenes (A,B,C)
Statement II : In Human skin colour AABBCC show darkest skin colour expression and aabbcc produce Lightest skin colour expression.
In the light of above statements choose most appropriate answer from the options given below.

(1) Both statements are true.
(2) Both statements are false
(3) Statement I is true and II is false
(4) Statement I is false and II is true

18) Which of the following is correct statement about sickle cell anaemia :-

(1) It is an autosome linked dominant trait that can be transmitted from parents to the off springs.
(2) This disease is controlled by 3 pair of alleles.
(3) Homozygous individuals with genotype Hb^AHb^A show the disease phenotype.
(4) The defect is caused by the substitution of glutamic acid by valine at the 6th position of the β-globin chain of the hemoglobin molecule.

19) A pedigree is shown below for a disease that is autosomal dominant:



Find out genotype of individual number 3, 4 and 5 respectively?

(1) Aa, Aa, Aa
(2) Aa, aa, Aa
(3) Aa, aa, aa
(4) aa, AA, Aa

20) Match the column-I with column-II :

Column - I		Column - II	
(A)	Determine genotype of tall plant	(i)	Test cross

(B)	Gene which code for a pair of contrasting traits	(ii)	Allele
(C)	Continuous self pollination	(iii)	Independent assortment
(D)	Independent gene pair	(iv)	True breeding line

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-iii, C-iv, D-i

(3) A-i, B-ii, C-iii, D-iv

(4) A-iv, B-iii, C-ii, D-i

21) Select the correct statement.

(1) To determine the genotype of tall plant at F_2, Mendel crossed the tall plant from F_2 with a dwarf plant.

(2) Factor never occurs in pair.

(3) Genes which code for different characters are called alleles.

(4) Mendel selected 7 true-breeding pea varieties.

22) In maize coloured endosperm (C) is dominant over colourless (c) and full endosperm (R) is dominant over shrunken (r). When a dihybrid of f1 generation was test crossed it produced four phenotypes in the following percentage.

Coloured and full = 40%
Coloured and shrunken = 10%
Colourless and full = 8%
Colourless and shrunken = 42%

From these data what would be the distance between the two non allelic genes :-

(1) 40 cM

(2) 42 cM

(3) 18 cM

(4) 10 cM

23) Transcriptionally active chromatin is termed as :-

(1) Euchromatin

(2) Neochromatin

(3) Heterochromatin

(4) Prochromatin

24) Fully processed hn RNA is called :-

(1) Sn RNA

(2) t RNA

(3) m RNA

(4) r RNA

25) Failure of segregation of chromatids during cell division cycle leads to :-

(1) Polyploidy
(2) Aneuploidy
(3) Chromosomal abberrations
(4) Mendelian disorders

26) In the first phase itself amino acids are activated in the presence of ATP and linked to their cognate tRNA. The process is commonly called as:

(1) Charging of tRNA
(2) Aminoacylation of tRNA
(3) Loading of tRNA
(4) All of the above

27) Splicing represents the Dominance of which world ?

(1) Protein
(2) DNA
(3) RNA
(4) Both (2) & (3)

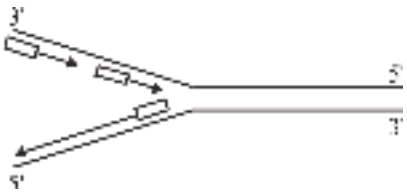
28) In capping an unusual nucleotide is added to the 5′-end of hnRNA which is :-

(1) Methyl guanosine triphosphate
(2) Methyl guanosine diphosphate
(3) Methyl guanosine monophosphate
(4) Guanosine triphosphate

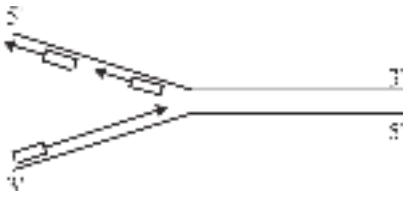
29) Who is credited for developing method for determination of amino acid sequence in proteins ?

(1) Frederick Griffith
(2) Har Gobind Khorana
(3) Frederick Sanger
(4) Severo Ochoa

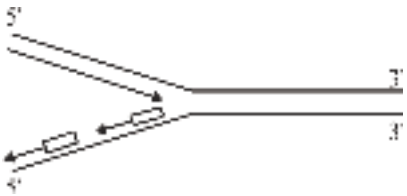
30) Which of the following diagram correctly represents the replication of DNA :-

(1) 

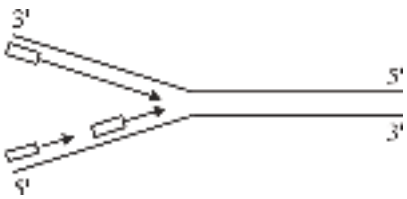
(2)



(3)



(4)



31) N-glycosidic linkage in a polynucleotide chain joins?

(1) A nucleotide to a ribose sugar

(2) A nitrogenous base to pentose sugar

(3) A nucleoside to a pentose sugar

(4) A phosphate group to a pentose sugar

32) Which is the primary step for regulation of gene expression, in eukaryotes-

(1) Transport of m-RNA from nucleus to the cytoplasm.

(2) Translational level

(3) Processing level

(4) Transcriptional level

33) Select the **incorrect** statement(s) :-
(A) Six-codons do not code for any amino acid.
(B) Codons are read in m-RNA in a contiguous fashion.
(C) Three codons function as stop codons.
(D) The initiator codon AUG codes for methionine.

(1) A and B

(2) A and D

(3) A and C

(4) Only A

34) Find out incorrect statement with respect to DNA structure.

(1) Two polynucleotide chains in a DNA are linked through hydrogen bonds.

(2) Two adjacent nucleotides in each polynucleotide chain of DNA, are linked through 3' - 5' phosphodiester linkage.

(3) Guanine is bonded with cytosine with three hydrogen bonds while adenine forms two hydrogen bonds with 5-methyl uracil.

(4) The pitch of DNA helix is 34 nm and there are roughly 10 bp in each turn, hence the distance between two adjacent bps in a helix is approximately 3.4 nm

35) Read the following statements-
(i) DNA from every tissue from an individual shows same degree of polymorphism.
(ii) The methodologies of HGP involves only one major approach of sequencing DNA of genome. (iii) In lac operon, y-gene codes for permease which increases permeability of cell membrane to β-galactosides. How many statements are correct?

(1) One

(2) Two

(3) Three

(4) None

36) Match the column correctly :-

	Column-I		**Column-II**
(i)	Hershey & Chase	(a)	Deciphering of genetic code
(ii)	Avery, Macleod and McCarty	(b)	Genetic material in bacteriophage is DNA
(iii)	Marshall Nirenberg	(c)	Concept of adapter molecule
(iv)	Francis Crick	(d)	In transforming principle of Griffith substance was DNA

(1) (i)-(a), (ii)-(d), (iii)-(b), (iv)-(c)

(2) (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)

(3) (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)

(4) (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a)

37) Find the correct match from the following table?

	Column-I	**Column-II**	**Column-III**
(i)	AUG	Start codon	Code for methionine amino acid
(ii)	GGG	First discovered codon	Code for phenyl alanine
(iii)	UGG	Termination codon	Do Not code for any amino acid

(1) i only

(2) i, ii

(3) i, iii

(4) ii, iii

38) Match List-I with List-II and select the correct answer using the code given below in the lists.

	List-I	List-II
a	Ligase	Joins short segments of DNA together.
b	DNA polymerase	Cuts protein at specific sequence
c	Helicase	Breaks the hydrogen bonds between complementary pairs during DNA replication.

(1) a, b and c are correct

(2) a and b are correct but c is false

(3) a is correct but b and c are false

(4) a and c are correct but b is false

39) **Assertion (A) :** The lac operon consists of polycistronic structural gene.
Reason (R) : After successful transcription a single mRNA produce three enzymes after translation in lac operon.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

40) Given below are two statements :
Statement I : DNA polymerases catalyse polymerisation only in one direction, that is 5' → 3'.
Statement II : During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.
In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

41) If in a DNA segment there are 100 base pairs, the number of thymine nitrogen bases in this segment is 46 then how many hydrogen bonds are present between both the strands of this DNA segment ?

(1) 200

(2) 254

(3) 246

(4) 218

42) **Statement-I :** The presence of introns is reminiscent of antiquity, and the process of splicing

represents the dominance of RNA-world.
Statement-II : The genetic code is nearly universal for example, from bacteria to human UUU would code for phenylalanine (phe).
Read the above statements and choose the correct answer ?

(1) Statement-I is correct and Statement-II are incorrect.

(2) Statement-I is incorrect and Statement-II are correct.

(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

43) Select the correct sequence of steps during DNA fingerprinting
a-Digestion; b-Isolation ; c-Autoradiography ; d-Blotting; e-Hybridisation; f-Electrophoresis

(1) a → b → c → d → e →f

(2) b → a → f → d → e → c

(3) c → d → a → e → f → b

(4) d → f → e → a → b → c

44) Match the column:

Column-I		**Column-II**	
(A)	Triplet	(i)	One codon codes for only one amino acid
(B)	Non-ambiguity	(ii)	One amino acid is coded by more than one codon
(C)	Universality	(iii)	A codon is made up of three N_2 bases.
(D)	Degeneracy or redundancy	(iv)	From bacteria to human beings AUG codes for methionine

(1) A = (iii), B = (ii), C = (iv), D = (i)

(2) A = (iii), B = (i), C = (iv), D = (ii)

(3) A = (iv), B = (i), C = (ii), D = (iii)

(4) A = (iii), B = (iv), C = (i), D = (ii)

45) **Assertion (A) :** When ribosome proceeds to the elongation phase of translation, complexes compound of an amino acids linked to tRNA, sequentially bind to the appropriate anticodon in mRNA by forming complementary base pairs with the tRNA codon.
Reason (R) : The rule of base pairing and complementarily is applicable in the process of DNA replication, transcription and translation.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

46) In agarose gel electrophoresis :-

(1) RNA migrates towards negative electrode
(2) Larger molecules migrate faster than smaller molecules
(3) Smaller molecules migrate faster than larger molecules
(4) DNA do not show migration

47) After completing the transfer of a desired DNA in vector which was having 'Lac Z' gene as marker gene, the recombinant colonies should :-

(1) Give blue colour
(2) Not give blue colour
(3) Have active galactosidase enzyme
(4) Both (2) & (3)

48) If we ligate a desired gene at Pvul site in pBR 322. then recombinants will:

(1) loose the tetracycline resistance
(2) not grow on tetracycline
(3) grow on ampicillin
(4) not grow on ampicillin

49) Plasmid used to construct the first recombinant DNA was Isolated from which bacterium?

(1) *E. coli*
(2) *Salmonella typhimurium*
(3) *Agrobacterium tumefaciens*
(4) *Thermus aquaticus*

50) Due to ampicillin resistance gene, one is able to select a transformed cell in the presence of ampicillin. The ampicillin resistance gene in this case is called :

(1) Recombinant gene
(2) Selectable marker
(3) Origin of replication
(4) Recognition site

51) In EcoRI (restriction endonuclease) the letter 'R' is derived from :

(1) Genus of bacteria
(2) Species of bacteria
(3) Strain of bacteria
(4) Class of bacteria

52) Which compound is used to stain DNA in Agarose Gel ?

(1) Ethidium bromide

(2) $NaNO_3$

(3) Ammonium nitrate

(4) Polyethylene glycol

53) **Assertion (A) :** Restriction endonucleases are called molecular scissors.
Reason (R) : They recognize specific palindromic sequences and cleaves the DNA at sugar-phosphate backbone.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and the Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is false.

(4) Assertion is false but the Reason is true.

54) Identify the incorrect statement related to restriction enzymes ?

(1) Hind II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs.

(2) In EcoRI, the letter 'R' is derived from the name of strain.

(3) Unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant vector molecule cannot be created.

(4) The first restriction endonuclease was Hind III.

55) Identify the correct statement related to gel electrophoresis ?

(1) Fragments of DNA moves towards cathode.

(2) The DNA fragments separate according to their size through sieving effect provided by the agarose gel.

(3) Separated DNA fragments can be directly seen under UV radiation.

(4) Sample of DNA is loaded at anode end of the gel.

56)

	Column-(A)		**Column-(B)**
(A)	Plasmid	(i)	Small circular DNA in bacteria
(B)	Bacteriophage	(ii)	Virus used to deliver gene
(C)	Ti plasmid	(iii)	Deliver gene into animal cell
(D)	Retrovirus	(iv)	Used in plant genetic engineering

(1) A-i, B-ii, C-iii, D-iv

(2) A-i, B-ii, C-iv, D-iii

(3) A-i, B-iv, C-iii, D-ii

(4) A-ii, B-i, C-iv, D-iii

57) Given below are two statements :

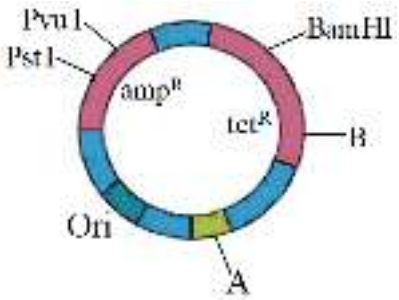
Statement-I : Each restriction endonuclease functions by inspecting the length of DNA sequences.
Statement-II : More than 900 restriction enzymes have been isolated from 230 strains of bacteria.
In the light of the above statements choose correct answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.



58) In the following diagram find out A and B :

(1) A ⇒ rop, B ⇒ Sal I

(2) A ⇒ Sal I, B ⇒ rop

(3) A ⇒ EcoRI, B ⇒ rop

(4) A ⇒ rop, B ⇒ Cla I

59) Select the correct sequence of PCR :-

(1) Denaturation → Annealing → Extension

(2) Annealing → Denaturation → Extension

(3) Extension → Denaturation → Annealing

(4) Denaturation → Extension → Annealing

60) Which of the following enzyme(s) are not involved in isolation of the genetic material (DNA) in process of recombinant DNA technology ?
A. Lysozyme
B. Cellulase
C. DNase
D. RNase
E. DNA ligase

(1) A and B only

(2) C and E only

(3) C and D only

(4) D and E only

61) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): If one wants to recover many copies of the target DNA, it should be cloned in a vector whose origin support high copy number.
Reason (R): The origin of replication (Ori) controls how many copies of a plasmid are made inside a

host cell so if a vector has a high-copy-number, it replicates many times per cell, producing hundreds of plasmid copies.
Choose the correct answer from the options given below:

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

62) Identify incorrect matches :

(a)	Ori	gene cloning
(b)	Plasmid	Circular, main chromosomal DNA
(c)	Pvu II	Site within rop
(d)	Plant cell	Biolistic
(e)	Primers	Large chemically synthesized polynucleotide
(f)	Agarose gel	Artificial polymer

(1) b, c

(2) a, b, c, e

(3) b, c, e, f

(4) b, e, f

63) *Meloidegyne incognitia* infects :

(1) Leaf of potato

(2) Stem of Soyabean

(3) Root of tobacco

(4) Flower of tomato

64) Over 95 percent of all existing transgenic animals are :

(1) Pigs

(2) Cows

(3) Fish

(4) Mice

65)

In 1997, an American company got patent rights for which indian crop illegally :

(1) Rice

(2) Wheat

(3) Maize

(4) Turmeric

66) CryIAc endotoxin obtained from *Bacillus thuringiensis* are effective against.

(1) Nematodes
(2) Bollworms
(3) Mosquitoes
(4) Flies

67) The laws and rules to prevent unauthorised exploitation of bio-resources are termed as -

(1) Biopatenting
(2) Bioethics
(3) Bioengineering
(4) Biopiracy

68) What triggers activation of protoxin to active Bt toxin of *Bacillus thuringiensis* in boll worm?

(1) Body temperature
(2) Moist surface of midgut
(3) Alkaline pH of gut
(4) Acidic pH of stomach

69) The first clinical gene therapy was given for treating

(1) Rheumatoid arthritis
(2) Adenosine deaminase deficiency
(3) Diabetes
(4) Chicken pox

70) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : One can grow meristem in vitro to obtain virus free plants.
Reason (R) : Even if the plant is infected with a virus, the meristem (apical and axillary) is free of virus.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both assertion and Reason are true and reason is correct explanation of assertion.
(2) Both assertion and Reason are true but reason is not correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

71) Which of the following statement is incorrect regarding genetically engineered insulin ?

(1) The main challenge for production of insulin using rDNA technique was getting insulin assembled into a mature form.
(2) Eli Lilly prepared two DNA sequences corresponding to A and B chains.

(3) Eli Lilly produces chains A and B separately.

(4) C-peptide was removed after creating disulfide bond.

72) Match the column-I with column-II.

	Column-I		**Column-II**
(A)	Insulin	(i)	Use to treat emphysema
(B)	Gene therapy	(ii)	Control blood glucose level
(C)	ELISA	(iii)	To treat some heriditary disease
(D)	α-1-Antitrypsin	(iv)	Based on antigen-antibody interaction

(1) A-ii, B-iv, C-iii, D-i

(2) A-iii, B-ii, C-iv, D-i

(3) A-i, B-iii, C-iv, D-ii

(4) A-ii, B-iii, C-iv, D-i

73) **Statement-I :-** Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs.
Statement-II :- In 1981, Eli Lilly an American company prepared human insulin from r-DNA technique.

(1) **Statement I** incorrect

(2) **Statement II** correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) Both **Statement I** and **Statement II** are correct.

74) Given below are the two statements :-
Statement-I :- Serum and urine analysis are used in early detection of disease.
Statement-II :- Probe is double stranded DNA tagged with radio active molecule.
Select the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Only statement I is incorrect

(4) Only statement II is correct

75) **Assertion (A) :** In detection of gene mutation using radioactive probe the clone having mutated gene does not appear on photographic film.
Reason (R) : The probe will not have complementarity with the mutated gene.

(1) Both A and R are correct and R is correct explanation of A.

(2) Both A and R are correct but R is not correct explanation of A.

(3) A is correct but R is incorrect.

(4) Both A and R are incorrect.

76) Match the Column I and II :

	Column I		Column II
A	Rosie	i	Emphysema
B	Polio vaccine	ii	Cystic fibrosis
C	Alpha-1-antitrypsin	iii	Alpha lactalbumin
D	Transgenic models	iv	Transgenic mice

(1) A-iii, B-iv, C-i, D-ii

(2) A-iv, B-i, C-ii, D-iii

(3) A-ii, B-iv, C-i, D-iii

(4) A-iii, B-ii, C-iv, D-i

77) Two statements are given below :
Statement-I :- *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). *B. thuringiensis* forms protein crystals during a particular phase of their growth. These crystals contain a toxic insecticidal protein.
Statement-II :- Actually, the Bt toxin protein exist as inactive protoxins but once an insect ingest the inactive toxin, it is converted into an active form of toxin due to the acidic pH of the gut which solubilise the crystals. The activated toxin binds to the surface of midgut epithelial cells and create pores that cause cell swelling and lysis and eventually cause death of the insect.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

78) Read the given following statements :-
(a) The recombinant therapeutics do not induce unwanted immunological responses.
(b) At present, about 20 recombinant therapeutics have been approved for human-use the world over
(c) Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs
(d) In mammals, including humans, insulin in synthesised as a pro-hormone
(e) ADA enzyme is crucial for the immune system to function
Choose the correct answer from the options given below :

(1) (b), (d), (e) only

(2) (a), (b), (c), (d), (e)

(3) (a), (c), (d), (e) only

(4) (a), (c) only

79) Wine and beer are produced by fermentation of sugar using___.

(1) *Rhizopus*

(2) *Saccharomyces cerevisiae*

(3) *Lactobacillus*

(4) *Aspergillus niger*

80) Flocs formed during secondary sewage treatment are a mixture of bacteria and_____.

(1) Fungi
(2) Viruses
(3) Protozoa
(4) Algae

81) Select the **correct** biocontrol agents

(1) Dragonflies and yeast
(2) Baculoviruses and aphids
(3) *Bacillus thuringiensis* and mosquitoes
(4) *Trichoderma* and *Nucleopolyhedrovirus*

82) Which of the following bacteria is present in the rumen of cattle ?

(1) *Rhizobium*
(2) *Azotobacter*
(3) *Methanobacterium*
(4) *Clostridium*

83) A biocontrol agent used for pest butterfly caterpillars is :

(1) *Trichoderma*
(2) *Bacillus thuringiensis*
(3) *Pseudomonas*
(4) *Rhizobium*

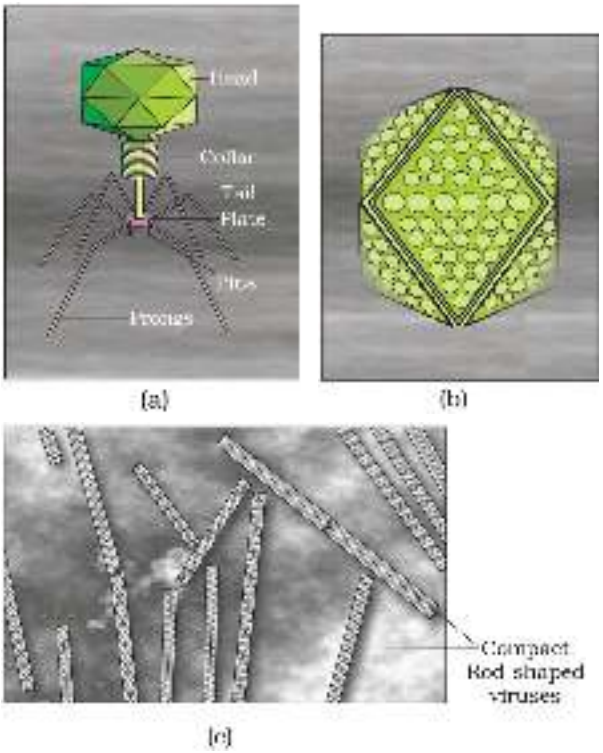
84) The majority of baculoviruses used as biological control agent are in the genus :-

(1) NPV
(2) CPV
(3) RPV
(4) RPV

85) **Assertion :-** 'Cyclosporin-A' is used as an immunosuppressive agent in organ-transplant patients.
Reason :- 'Cyclosporin-A' is produced by the fungus *Trichoderma polysporum*.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but the Reason is True.

86) Identify the microbes in the given diagrams and select the correct option :



(1) a is adenovirus

(2) c is tobacco mosaic virus

(3) b is bacteriophage

(4) c is adenovirus

87) Which of the following statement is incorrect ?

(1) In paddy fields, cyanobacteria serve as an important biofertiliser.

(2) Baculoviruses are pathogens that attack insects and other arthropods.

(3) Dragonflies are useful to get rid of aphids.

(4) Trichoderma species are effective biocontrol agents of several plant pathogens.

88)

Column-A		**Column-B**	
(A)	*Sacchromyces cerevisiae*	(i)	Cyclosporin-A
(B)	*Aspergillus*	(ii)	Statin
(C)	*Monascus purpureus*	(iii)	Ethanol
(D)	*Trichoderrma polysporum*	(iv)	Citric acid

(1) A-iii, B-ii, C-iv, D-i

(2) A-iii, B-i, C-ii, D-iv

(3) A-iii, B-iv, C-ii, D-i

(4) A-iv, B-iii, C-ii, D-i

89) Given below are the two statements:
Statement-I : Bioactive molecule cyclosporin-A produced by fungus *Trichoderma polysporum* is used as an immune suppressive agent in organ transplant patients.
Statement-II : Many members of genus glomus form mycorrhiza.

(1) Statement-I is correct but statement-II is incorrect.

(2) Both Statement-I and statement-II are correct.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both Statement-I and statement-II are incorrect.

90) Consider the following statements :
(i) Till date, no man-made technology has been able to rival the microbial treatment of sewage.
(ii) The effluent from the primary treatment plant is generally released into natural water bodies like rivers & streams.
(iii) The major part of the activated sludge is pumped into large tanks called anaerobic sludge digesters.
(iv) The sewage water is treated till the BOD is increase.
(v) The BOD test measures the rate of uptake of oxygen by microorganisms in a sample of water.
(vi) Bacteria which grow anaerobically on cellulosic material, produce large amount of methane along with CO_2 and H_2.
Which of the following option have all the incorrect ones ?

(1) Only (i) and (ii)

(2) Only (iii) and (v)

(3) Only (ii) and (iv)

(4) Only (iv) and (vi)

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	4	1	3	2	3	1	2	1	3	2	2	2	2	2	3	4	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	2	2	4	1	4	3	1	1	3	2	1	2	4	4	4	2	4	4
Q.	41	42	43	44	45															
A.	1	1	4	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	2	3	1	3	4	3	3	4	2	4	4	2	1	3	4	2	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	2	4	3	1	3	3	3	4	1	2	4	1	1	2	3	2	1	2
Q.	86	87	88	89	90															
A.	2	1	3	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	1	3	2	3	2	4	1	3	4	3	1	3	3	1	1	4	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	1	3	2	4	3	1	3	2	2	4	4	4	2	2	1	4	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	2	2	4	3	2	4	2	2	3	1	1	4	2	2	1	1	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	3	4	1	2	1	3	2	1	4	4	3	2	1	1	3	3	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	2	1	2	2	3	3	2	3										

SOLUTIONS

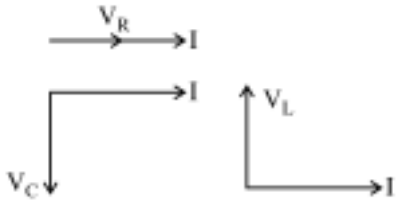
PHYSICS

1)

Charge flown = $\frac{\Delta\phi}{R}$

Area of I – t curve = $\frac{\text{change in flux}}{R}$

⇒ change in flux = $\frac{1}{2} \times 2 \times 0.1 \times 20 = 2\,\text{wb}$

2) 

3)

$X_C > X_L$. Hence, current will lead the voltage.

$Z = \sqrt{R^2 + (X_C - X_L)^2}$

$= 10\sqrt{2}\,\Omega$

$\Rightarrow I_{rms} = \frac{V_{rms}}{Z}$

$= \frac{400/\sqrt{2}}{10\sqrt{2}} = 20$ A

$\cos\phi = \frac{R}{Z} = \frac{1}{\sqrt{2}}$

$V_R = I_{rms} R = (20)(10) = 200$ V

4)

$V = \frac{4V_0}{T}t$

$V_{rms} = \sqrt{\frac{\int_0^{T/4} V^2 dt}{T/4 - 0}} = \frac{V_0}{\sqrt{3}}$

5) $f = \frac{1}{2\pi\sqrt{LC}}$

$f' = \frac{1}{2\pi\sqrt{L(KC)}}$

$f' = f - 25 = 125 - 25 = 100$ KHz

$\left(\frac{f}{f'}\right)^2 = K \Rightarrow K = 1.56$

6) $\cos\phi = \frac{1}{\sqrt{2}}$

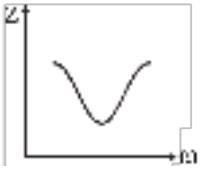
$\therefore \phi = \frac{\pi}{4}$

$\therefore |X_L - X_C| = R$

$|X_L - X_C| = 10$

$|0.1 \times 100 - X_C| = 10$

$\therefore X_C = 0$

7) 

$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

if $\omega \to$ Low $\quad Z \to \frac{1}{\omega C}$

$\omega \to$ High $\quad Z \to \omega L$

8) using KCL, $I_3 = I_1 + I_2$

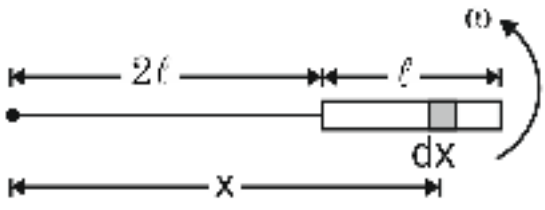
$I_3 = 3A \sin \omega t + 4A \cos \omega t$

$I_3 = R \sin(\omega t + \phi)$

where $R = \sqrt{(3A)^2 + (4A)^2} = 5A$

$\tan\theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$

$I = 5 \sin(\omega t + 53°)$

9) 

Consider an element of length dx of the rod at a distance x from the fixed end of the string. e.m.f. induced in the element is $de = B(\omega x)dx$

Hence, e.m.f. induced across the ends of the rod is $e = \int_{2\ell}^{3\ell} B\omega x dx = B\omega\left[\frac{x^2}{2}\right]_{2\ell}^{3\ell}$

$= \frac{B\omega[(3\ell)^2 - (2\ell)^2]}{2} = \frac{5B\omega\ell^2}{2}$ Hence, correct answer is (b).

10) $\oint \vec{E}.\vec{d\ell} = -\frac{d\phi}{dt}$

$E \cdot 2\pi R = -\frac{B_0 \pi a^2}{\Delta t}$

$\tau = Fr_{\perp} = (\lambda 2\pi R)\, ER$

$\tau \Delta t = I\omega - 0$

$\lambda 2\pi R^2 E \Delta t = MR^2\omega$

$\lambda R\left(\frac{-B_0 \pi a^2}{MR^2}\right) = \omega$

$\Rightarrow \omega = \frac{-B_0 \pi a^2 \lambda}{MR}\hat{k}$

11)

$P_{inst} = VI$

if $V = V_0 \sin \omega t$ then

$I = I_0 \sin(\omega t - \pi/2)$

for inductor only

$P = V_0 \sin \omega t\, I_0 \sin(\omega t - \pi/2)$

$P_{inst} = -(V^0 \sin \omega t)\left\{\frac{V_0 \cos \omega t}{\omega L}\right\} = \frac{-V_0^2}{2\omega L}\sin 2\omega t$

12)

The current through an RL circuit after the switch is turned off decays according to :

$I(t) = I_0 e^{-t/\lambda}$

where $\tau = \frac{L}{R}$ is the time constant of the circuit

As the current decreases over time due to the inductor opposition to sudden changes in current both bulbs experiences a delay in their brightness fading. Since there s' an inductor in the circuit neither bulb will die out instantly.

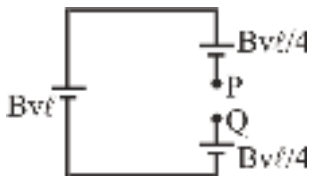
Hence, option (2) is correct.

13) Due to magnetic field, wire will experience an upward force $F = Bi\ell = B\left(\frac{Bv\ell}{R}\right)\ell$

$\Rightarrow F = \frac{B^2 v \ell^2}{R}$ If wire slides down with constant velocity then

$F = mg \Rightarrow \frac{B^2 v \ell^2}{R} = mg \Rightarrow v = \frac{mgR}{B^2\ell^2}$

14)



Using KVL

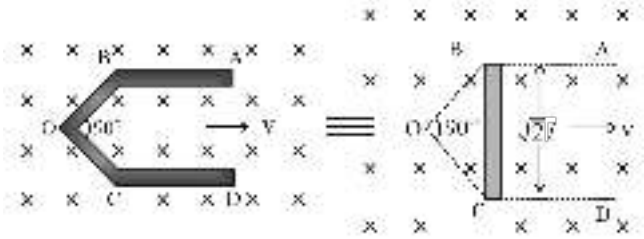
$V_P + \frac{B v \ell}{4} - B v \ell + \frac{B v \ell}{4} = V_Q$

$V_P - V_Q = \frac{B v \ell}{2}$

$V_P - V_Q$ is positive so $V_P > V_Q$

15) $V_A - V_B = \frac{Ldi}{dt}$

16) There is no induced e.m.f. in the part AB and CD because they are moving along their length while e.m.f. induced between B and C i.e., between A and D can be calculated as follows.



Induced e.m.f. between B and C = Induced e.m.f. between.
A and B = Bv ($\sqrt{2}$l)
= 1 × 1 × 1 × $\sqrt{2}$ = 1.41 volt.

17) The field produced due to induced current is into the paper. So, current in outer loop is clockwise and in inner loop is anticlockwise. Hence, current will flow from A to B and D to C.

18) $\eta = \frac{P_{out}}{P_{in}} \times 100$, where $P_{in} = V_P I_P$

$= \left(\frac{100}{220 \times 0.5}\right) \times 100 = 90\%$

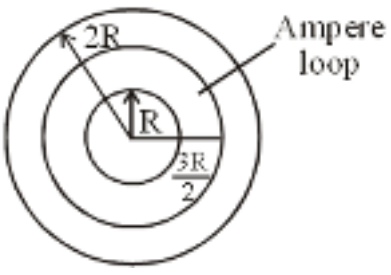
19)

$e = (\vec{v} \times \vec{B}) . \ell$

$e = [\hat{i} \times (3\hat{i} + 4\hat{j} + 5\hat{k})] . 5\hat{j}$

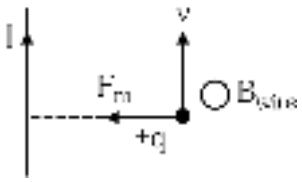
⇒ e = 25 volt

20)



$B = \frac{\mu_0 I}{2\pi r} \frac{(r^2 - a^2)}{(b^2 - a^2)}$

$$B = \frac{\mu_0 I}{2\pi \frac{3R}{2}} \frac{\left(\frac{9R^2}{4} - R^2\right)}{(4R^2 - R^2)} = \frac{5\mu_0 I}{36\pi R}$$

21) 

by night hand palm rule; force on charge +Q is opposite to OX.

22) $B = \frac{\mu_0 i}{2\pi r} = \frac{2 \times 10^{-7} \times 50}{2.5}$
$= 4 \times 10^{-6}$ T

23) $a = \frac{f}{m} = \frac{Bi\ell}{m}$
$= \frac{2 \times 2 \times 3 \times 10^{-2}}{10 \times 10^{-3}} = 12 \text{ ms}^{-2}$
here effective length = AC = 3 cm

24)

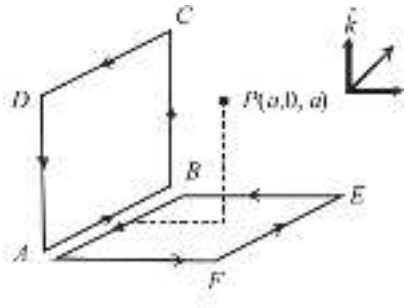
$B^0 = \frac{\sqrt{2}\mu_0 I_1}{\pi d} - \frac{\sqrt{2}\mu_0 I_2}{\pi d}$
$= \frac{\sqrt{2}\mu_0}{\pi d}(I_1 - I_2)$
$= \frac{\sqrt{2}\mu_0}{\pi d}\left(\frac{2I}{3} - \frac{I}{3}\right) = \frac{\sqrt{2}\mu_0 I}{3\pi d}$

25) The energy of a charged particle moving in magnetic field remains constant because the magnetic field does not do any work. Therefore kinetic energy is constant i.e. u = v.
The force on electron will act along negative
y-axis initially. The electron will undergo circular motion in clockwise direction and emerge out the field. So $y < 0$.

26) Since, magnetic field at the centre of an arc is equal to
$B = \frac{\mu_0 I}{4\pi r}\theta$
Hence, net $B = \frac{\mu_0 I}{4\pi}\left[\frac{1}{r} - \frac{1}{2r} + \frac{1}{3r}\right]\theta \Rightarrow \frac{5\mu_0 I\theta}{24\pi r}$

27) The magnetic field at due to the loop is equal to the vector sum of the magnetic fields produced by loops ABCDA and AFEBA as shown in the figure.
Magnetic field due to loop ABCDA will be along $\hat{i}$ and due to loop AFEBA, along $\hat{k}$. Magnitude of magnetic field due to both the loops will be equal.

Therefore, direction of resultant magnetic field at P will be $\frac{1}{\sqrt{2}}(\hat{i}+\hat{k})$.



28) $B_{AB} = \frac{\mu_0 I}{4\pi(OC)}[2\sin\theta]$

But OC = r cosθ or $B_{AB} = \frac{\mu_0 I}{2\pi r}\tan\theta$

Magnetic field due to circular portion,

$B_{AB} = \frac{\mu_0 I}{2r}\left(\frac{2\pi-2\theta}{2\pi}\right) = \frac{\mu_0 I}{2\pi r}(\pi-\theta)$

Total magnetic field

$= \frac{\mu_0 I}{2\pi r}\tan\theta + \frac{\mu_0 I}{2\pi r}(\pi-\theta) = \frac{\mu_0 I}{2\pi r}[\tan\theta+\pi-\theta]$

29) $H = \frac{\mu_0 ni}{\mu_0} = ni = \frac{Ni}{\ell}$

$i = \frac{H\ell}{N}$

30) $\mu_r = 1 + \chi_m$

For diamagnetic

$-1 \le \chi_m < 0$

For paramagnetic

$0 < \chi_m < 1$

For ferromagnetic

$\chi_m >> 1$

31)

$\mu = \frac{B}{H} = \frac{\phi}{AH} = \frac{4\times10^{-4}}{4\times10^{-4}\times1600}$

$= 0.625 \times 10^{-3}$

32)

$M = \left(\frac{q}{2m}\right)L \Rightarrow M = \left(\frac{q}{2m}\right)I\omega$

$$\Rightarrow \frac{M}{I} = \frac{q}{2m}\, 2\pi f \Rightarrow \frac{M}{I} = \frac{q\pi f}{m}$$

33)

Magnetic field at the center of a circular orbit:

$$B = \frac{\mu_0 I}{2r}, I = \frac{e\omega}{2\pi}$$

Diameter of orbit = 10^{-10}m

$$r = \frac{1}{2} \times 10^{-10} = 5 \times 10^{-11}\ m$$

Magnetic field at the center:

$$B = \frac{\mu_0}{2r} \cdot \frac{e\omega}{2\pi} = \frac{\mu_0 e\omega}{4\pi r}$$

Using $\mu_0 = 4\pi \times 10^{-7}$

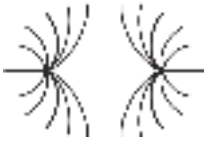
$$B = \frac{\left(4\pi \times 10^{-7}\right) e\omega}{4\pi r} = \frac{e\omega}{r} \times 10^{-7}$$

Given $B = 14\ Wb\ m^{-2}$:

$$14 = \frac{\left(1.6 \times 10^{-19}\right) \omega}{5 \times 10^{-11}} \times 10^{-7}$$

$$\omega = \frac{7}{1.6} \times 10^{16} rad/s \approx 4.375 \times 10^{16} rad/s.$$

34) The iron filing shall be repelled at both the north poles on paper. The pattern will be.



35) Magnetic field lines due to current carrying solenoid produces a magnetic field that is nearly identical to that of bar magnet.

36) $B = \mu^0 \mu_r \left(\frac{N}{2\pi r}\right) i$

$$1 = 4\pi \times 10^{-7} \times \mu_r \times \frac{400}{40 \times 10^{-2}} \times 2$$

$$\mu_r = \frac{10^4}{8\pi} \simeq 400$$

37) $I_D = 1$ mA $= 10^{-3}$ A; C = 2 μF $= 2 \times 10^{-6}$ F

$$I_D = I_C = \frac{d}{dt}(CV) = C\frac{dV}{dt}$$

Therefore, $\frac{dV}{dt} = \frac{I_D}{C} = \frac{10^{-3}}{2 \times 10^{-6}} = 500\ Vs^{-1}$

Therefore, applying a varying potential difference of 500 Vs^{-1} would produce a displacement current of desired value.

38) $\vec{S} \Rightarrow \vec{E} \times \vec{B}\,\hat{j} \Rightarrow \hat{k} \times \hat{n}$
So $\hat{n} \Rightarrow \hat{i}$ i.e. Eastwards

39) Microwaves are used to cook food. Microwave oven is a domestic application of these waves.

40)

Statement-I is incorrect, Statement-II is correct.
$E = A \sin(kx - \omega t)$ and $u = \frac{1}{2}\varepsilon^0 E^2$
$u = \frac{1}{2}\varepsilon^0 (A \sin(kx - \omega t))^2 = \frac{1}{2}\varepsilon^0 A^2 \sin^2(kx - \omega t)$
$= \frac{1}{2}\varepsilon_0 A^2 \left[\frac{1 - \cos 2(kx - \omega t)}{2}\right]$
The energy density has double the frequency of the electromagnetic wave.

41) Given energy density = 50 W/m^2
Here, change in momentum $\Delta p = p_f - p_i$
$= \frac{-p_i}{4} - p_i = \frac{-5p_i}{4}$ $\quad p_i = \frac{U}{c} = \frac{50\ W/s}{3 \times 10^8 m/s}$
$F = \frac{\Delta p}{\Delta t} = \left|\frac{-5p_i}{4}\right| = \frac{5}{4} \times \frac{50}{3 \times 10^8}$
$= 20.8 \times 10^{-8}$ N
$\simeq 21 \times 10^{-8}$ N

42) P = 1000 W; r = 2m; η= 1.25%
Intensity, $I = \frac{1}{2}\varepsilon_0 C E_0^2$ .....(i)
$I = \frac{P}{area} \times \eta$ .....(ii)
From equation (i) and (ii),
$\frac{1000}{4\pi(2)^2} \times \frac{1.25}{100}$
$= \frac{1}{2} \times 8.854 \times 10^{-12} \times 3 \times 10^8 E_0^2$
E_0 = 13.7 V/m
As question,
x × 10^{-1} V/m = 13.7 V/m; x = 137

43) $G = \frac{RS}{R - S} = \frac{(4900)(98)}{4900 - 98} = 100\Omega$
Full scale current
$I_g = \frac{E}{R + G} = \frac{5}{4900 + 100} = 1$ mA

F.O.M = $\frac{I}{\theta} = \frac{1\,mA}{20}$ = 50 µA / division

44) $r = \frac{P}{qB} \Rightarrow P = qBr$
$P \propto q$ (B, r → same)
$\frac{p_p}{p_\alpha} = \frac{q_p}{q_\alpha} = \frac{1}{2} = \frac{e}{2e} = \frac{1}{2}$

45) $W = MB(\cos\theta_1 - \cos\theta_2)$
$\sqrt{3} = MB(\cos 0^\circ - \cos 60^\circ)$
$\sqrt{3} = \frac{MB}{2}$ ..... (i)
$\tau = MB\sin\theta$
$\tau = MB\sin 60^\circ = \sqrt{3}\frac{MB}{2}$ ..... (ii)
$\therefore \tau = \left(\sqrt{3}\right)\left(\sqrt{3}\right) = 3$ N-m

CHEMISTRY

46)

X → Na_2CrO_4 (Yellow solution)
Y → $PbCrO_4$ (Yellow ppt)

47)

All contain Cu as their one of the component.

48)

Colour of Fe in oxidising flame in borax bead test is yellow.

49)

$DMG^{\ominus}$ is used for detection of Ni^{+2}.

50)

MnF_6 does not exist due to steric repulsion.

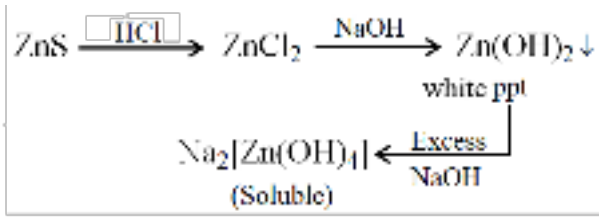
51)

$PbCl_2$ → White
$BaCrO_4$ → Yellow

MnS → Pink
NiS → Black

52)

Fe^{+3} gives blood red colour with KSCN.

53) 

54)

Group reagent for group IV
Cation ⇒ H_2S in presence of NH_4OH

55)

$[Ti(OC_6H_5)_4]$ is not an OMC because metal is not directly bonded with carbon.

56)

In Chromyl chloride test red orange vapours is obtained.

57)

H_2S will not give any colour with SNP because it is a weak electrolyte.

58)

$Cu^{+2} \xrightarrow{K_4[Fe(CN)_6]} Cu_2[Fe(CN)_6]$ (Chocolate brown ppt)

59)

Compound	**SI**
$[Co(NO_3)_3(NH_3)_3]$	2
$[CoBrCl(OX)_2]^{3-}$	3
$K_3[Fe(OX)_3]$	2
$[CoCl_2(en)_2]^{+}$	3

60)

Na_2HPO_4 is used for detection of Mg^{+2}

61)

If HNO_3 is not added Fe^{+2} will not change to Fe^{+3}

62)

NH_3 gives brown ppt with Nessler's reagent

63)

Cation	**Group reagent**
Pb^{+2}, Cu^{+2}	H_2S + HCl
Al^{+3}, Cr^{+3}	NH_4OH + NH_4Cl
Co^{+2}, Ni^{+2}	H_2S + NH_4OH
Ba^{+2}, Ca^{+2}	$(NH_4)_2CO_3$ + NH_4OH

64)

Test of NH_4^+ done in zero group

65)

$CoO.Al_2O_3$ → Thenard blue

66)

BeH_2(s), B_2H_6 ⇒ Incomplete octet
I_2Cl_6 ⇒ Hyper valent
$BeCl_2$(s) ⇒ Complete octet

67)

$H_3P_3O_9$ or $(HPO_3)_3$ is a cyclic oxy acid

68)

Purest allotrope of carbon ⇒ Fullerene
Thermodynamically most stable
Allotropes of carbon ⇒ graphite

69)

$CO_2 < SiO_2 < GeO_2 < SnO_2$ (Oxidising power)
$MnO_4^- > TcO_4^- > ReO_4^-$ (Oxidising power)
$CH_4 < SiH_4 < GeH_4 < SnH_4$ (Reducing power)

70)

Reducing power of group 15 increases on moving down the group.

71)

Single N-N bond is weaker than single P-P bond due to lp-lp repulsion.

72)

$X_2 \rightarrow I_2$
$Y_2 \rightarrow Br_2$ or I_2

73)

XeF_4 act as F^- acceptor with KF.



74)

Correct formula is $K_4[Ni(CN)_2(OX)_2]$

75)

$[V(CO)_6]^-$ has more synergic bonding and less (C-O) bond order

76)

Both the Statement are true.

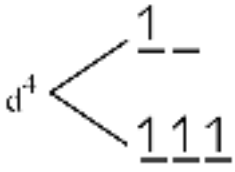
77)

$[Co(NH_3)_5Cl]Cl_2 \Rightarrow$ 1 : 2 electrolyte

78)

$[Ni(NH_3)_6]^{+2} \Rightarrow sp^3d^2$
$\Rightarrow$ Outer orbital complex

79)



CFSE = $-0.4\Delta_0 \times 3 + 0.6\Delta_0 \times 1$
$= -1.2\Delta_0 + 0.6\Delta_0$
$= -0.6\Delta_0$

80)

$Na_2[CdCl_4]$ is not coloured as Cd^{+2} is d^{10}

81)

→ Cr^{+2} is better reducing agent than Fe^{+2}
→ $E^{o}_{M^{3+}/M^{+2}}$ value is –ve for Cr but +ve for Fe

82)

X → MnO_2, Black
Y → K_2MnO_4, green
Z → $KMnO_4$

83)

Fe_3C and Mn_4N are interstitial compounds

84)

I_2O_5 is used for the estimation of CO.

85)

$[Fe(CO)_4]^{-2}$
Hybridisation → sp^3
Geometry → Tetrahedral

86)

SF_6 → gas
SeF_6 → gas
TeO_2 → oxidising agent
SeF_4 → liquid

87)

for $[Ir(\underline{H_2O})_6]^{+3}$
↓
SFL ⇒ $\Delta_0 > P$

88)

CuF does not exist
$2CuF \rightarrow CuF_2 + Cu$

89)

He < Ne < Ar < Kr < Xe
(Solubility in H_2O)

90)

$K_2Cr_2O_7 \xrightarrow{CO_2}$
(Orange) No colour change observed

BIOLOGY

91) NCERT-XII, Pg. # 61

92) NCERT-XII, Pg. # 67

93) NCERT-XII, Pg. # 60

94) NCERT-XII, Pg. # 58

95) NCERT-XII, Pg. # 72

96) NCERT-XII, Pg. # 62

97) NCERT-XII, Pg. # 74

98) NCERT-XII, Pg. # 70

99)

NCERT-XII, Pg. # 73

100) NCERT-XII, Pg. # 56

101)

NCERT-XII, Pg. # 69

102) NCERT-XII, Pg. # 56

103) NCERT-XII, Pg. # 73, 74, 75

104) NCERT-XII, Pg. # 67

105)

NCERT-XII, Pg. # 70

106) NCERT-XII, Pg. # 57

107)

NCERT-XII, Pg. # 69

108) NCERT-XII, Pg. # 74

109)

NCERT-XII, Pg. # 72, 73

110) NCERT-XII, Pg. # 56, 58, 66

111) NCERT-XII, Pg. # 56

112) NCERT-XII, Pg. # 67

113)

NCERT-XII, Pg. # 84

114) NCERT-XII, Pg. # 94

115) NCERT-XII, Pg. # 75

116) NCERT-XII, Pg. # 98

117) NCERT-XII, Pg. # 88

118) NCERT-XII, Pg. # 94

119) NCERT-XII, Pg. # 103

120)

NCERT-XII, Pg. # 91

121)

NCERT-XII, Pg. # 81

122) NCERT-XII, Pg. # 99

123)

NCERT-XII, Pg. # 96

124) NCERT-XII, Pg. # 81, 82

125) NCERT-XII, Pg. # 101, 104

126)

NCERT-XII, Pg. # 82, 84, 85, 96

127)

NCERT-XII, Pg. # 96

128)

NCERT-XII, Pg. # 90

129) NCERT-XII, Pg. # 100

130)

NCERT-XII, Pg. # 90

131)

NCERT-XII, Pg. # 81

132) NCERT-XII, Pg. # 95, 96

133)

NCERT-XII, Pg. # 105

134) NCERT-XII, Pg. # 96

135) NCERT-XII, Pg. # 99

136) NCERT-XII, Pg. # 198

137) Correct Answer: (2)

138) NCERT Pg.# 169

139) NCERT Pg.# 164

140) NCERT Pg.# 169

141) NCERT Pg.#195

142)

NCERT Pg.# 168

143) NCERT-XII, Pg. No. # 166

144) NCERT-XII, Pg.# 165, 166, 167 (E) ; 181 to 184 (H)

145) NCERT XII, Pg.# 168 (E), 184 (H)

146) NCERT Pg. # 168, 170

147) NCERT Pg. # 165, 166 (E), 182 (H)

148) NCERT Pg # 199

149) NCERT Pg # 202 (E), 222 (H)

150) NCERT-XII, Pg. # 171 (E), 188 (H)

151) NCERT Pg. # 168, 169

152) NCERT Page no. 164, 169, 17, 173

153)

NCERT Pg.# 180

154) NCERT Pg.# 181

155)

NCERT Pg.# 185

156)

NCERT XII, Pg # 209

157)

NCERT Pg.# 184-185

158)

NCERT Pg.# 179

159)

NCERT Pg.# 182

160) NCERT Pg. No. # 178

161) NCERT-XII, Pg. # 182 (E), 199 (H)

162) NCERT Pg. No. # 182-184

163) NCERT-XII, Pg. # 181

164) NCERT Page No. 182, 183 (E) 200 (H)

165) NCERT-XII → 183 (E)/ 200 (H)

166) XII NCERT Pg # 212-213

167) NCERT-XII, Pg. # 179

168) NCERT-XII, Pg. # 181 & 182

169) Saccharomyces cerevisiae

170) Correct option 1

171)

NCERT Pg.# 156, 157

172)

NCERT-XII, Pg. # 155

173)

NCERT Pg.# 157

174)

NCERT Pg.# 157

175) NCERT Pg. # 153 (E), 168 (H)

176)

NCERT Pg.# 150

177) NCERT-XII, Pg. # 157, 158 (E), 173, 174 (H)

178) NCERT (XI) Page No. 152, 153

179) NCERT Page no. 153, 158(E), 168, 174(H)

180) NCERT-XII, Pg # 154,155